Answer ALL Questions

1. Question is on renewables in power system planning

a) A national power system is optimally planned and has an existing generation mix shown in Table 1 with a minimum demand of 8.36 GW.

Table 1. Existing generation mix for national power system

Technology	Rated Capacity (GW)	Capacity Value	Cost (£/MWh)	GHG Emissions (gCO_2/kWh)
Gas	1	0.96	60	500
Nuclear	12	0.95	10	0

(i) What is the expected maximum output of the Gas and Nuclear generation and the maximum demand that can be served? [4]

Expected maximum output of Gas = 0.96 *1 = 0.96 GW
Expected maximum output of Nuclear = 0.95 * 12 = 11.4 GW
Maximum Demand = 0.96 + 11.4 = 12.36 GW

(ii) Assuming that the demand is uniformly distributed between the minimum and the maximum calculate the yearly electrical energy production, capacity factors for gas and nuclear, yearly emissions and emissions per kWh generated. [10]

Yearly energy production is an average of 10.36 GW all year = 24 * 365 * 10.36 = 90, 753.6 GWh

Between 8.36 and 12 GW of demand i.e., (12 – 8.36)/(12.36-8.36) * 100 % = 91 % of the time (uniform distribution) nuclear is the cheapest and can supply all the demand and an average output of 8.36 + (12 – 8.36)/2 = 10.18 GW.

Between 12 GW and 12.36 GW of demand i.e., 9 % of the time – nuclear is at its maximum all the time and gas need to generate at an average of (0.36)/2 GW = 0.18 GW. Not needed.

Capacity Factor Nuclear = (0.91 *10.18 + 0.09 * 12)/12 = 0.862

Nuclear Energy = Capacity Factor Nuclear * 12 * 8760 * 365 = 90, 611.7 GWh

Gas Energy = 90,753.6 – 90, 611.7 = 141.9 GWh

Another way is Gas Energy = 0.18 * 0.09 * 8760 = 141.9 GWh Not needed see above

Capacity Factor Gas = (141.9)/ (8760) = 0.162

Emissions = 141.9 *10^6 * 500 gCO_2 = 70.956 kT per year

Emissions per kWh = 70.956/90, 753.6 = 7.8 * 10^{-4} g/kWh

(iii) Typically, nuclear generation is inflexible. The power system has a seasonal variation in demand with the demand being on average lower in the summer than the winter by 3GW. Is nuclear generation in this power system flexible or inflexible? Please very briefly

justify your answer.

[5]

The power system demand varies from 8.36 to 12.36 (4 GW) over the year. If we do not know how much it varies daily, then the Nuclear must be assumed to be flexible. In this case there is a seasonal variation of 3GW so that brings the daily variation to be 1GW and so Gas of 1GW is probably ample to deal with all the demand variations so nuclear is not flexible.

b) The government has set a policy target of net zero without adding to the nuclear fleet. Wind and solar resource characteristic are given in Table 2 below, and it is assumed they are emissions free.

Table 2. Wind and solar resource characteristics

Technology	Capacity Factor (%)	Capacity Value for installed capacity 0 to 1 GW	Capacity Value for installed capacity 1 to 3 GW	Capacity Value for installed capacity greater than 3GW	LCOE (£/MWh)
Wind	40	0.35	0.25	0.15	40
Solar	10	0.05	0.03	0.01	30

(i) How much wind will need to be installed to replace the gas generation and maintain reliability? [2]

Wind Installed * Capacity Value = 1 * 0.96

Wind Installed = (1*0.96)/(0.15) = 6.4 GW

(ii) Assuming flexible nuclear estimate the wind curtailment. [3]

If Nuclear is flexible there should be little curtailment as the Maximum wind is well below the Minimum Demand. So zero curtailment.

(iii) How much solar will need to be installed to replace the gas generation and maintain reliability? [2]

Solar Installed * Capacity Value = 1 * 0.96

Solar Installed = (1*0.96)/(0.01) = 96 GW

(iv) Assuming flexible nuclear, estimate the solar curtailment. [3]

Simply looking at the Energy from solar for a year = 0.1 * 96 * 8670 = 84,096 GWh almost the same as is required to serve the load for an entire year for the whole year 90, 753.6 GWh. Solar will only generate during the day and with a capacity factor of 10 % the average output

during the day is 20 % of 96 GW = 19.2 GW. The average demand is 10.36 GW so almost 50 % of all solar energy is curtailed i.e. – 45 TWh. This is regardless of Nuclear being flexible.

(v) Which is the best option wind or solar and very briefly justify your answer. [4]

Wind's LCOE is higher (40) and has no curtailment but 50 % of all solar is curtailed making the LCOE of solar almost 2 times higher than stated (30) so 60. So wind should be built.

2. Question on risk in power system planning

a) Table 3 below contains the objective function value per scenario of a power system planning model under various plans (noted as PF1-PF5, Plan 1, Plan 2). In detail, PF1 is the perfect foresight plan for scenario 1, i.e., it has been obtained assuming only scenario 1 is plausible. PF2 to PF5 are obtained in the same manner. Plans 1 and 2 have been obtained using methods that consider multiple scenarios within a single optimization problem.

Table 3. Objective function value in £

		Plan						
		PF1	**PF2**	**PF3**	**PF4**	**PF5**	**Plan 1**	**Plan 2**
Scenarios	**S1**	5250	5260	5400	5710	6610	5520	5600
	S2	4280	4265	4340	4580	4820	4460	4520
	S3	3660	3475	3150	3195	3330	3180	3300
	S4	3660	3475	2870	2450	2520	2450	2450
	S5	3660	3475	2870	2380	1960	2450	2310

i) Draw the cumulative risk profiles of the two plans (Plan1, Plan2) assuming that the five scenarios are equiprobable. Please show each step for constructing the cumulative risk profiles and your calculations. [5]

The marks here will be based on:

- Correct labelling of axes: 0.5 mark for each axis
- Correct shape (Maximum value on y axis (1) and monotonically increasing curve): 1 mark
- Correct method for creating the curve but numerical error (2)
- Correct curves (+1)

Step 1: For each plan, we need to sort the scenarios in ascending order with respect to the objective function value.

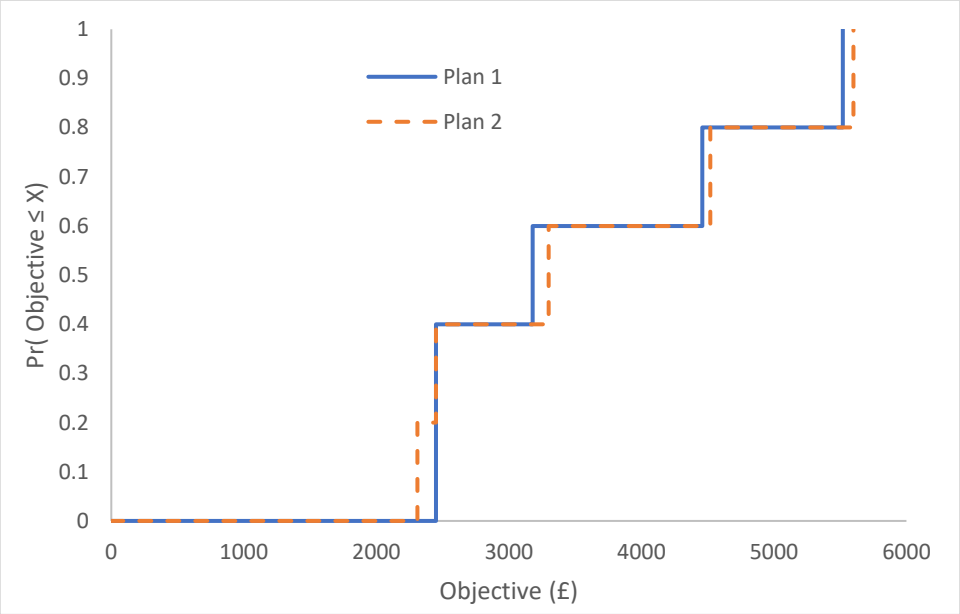
Plan 1 Objective function outcome (X)	
2450	
2450	
3180	
4460	
5520	

Plan 2 Objective function outcome (X)	
2310	
2450	
3300	
4520	
5600	

Step 2: Calculate cumulative probabilities.

Plan 1 Objective function outcome (X)	Pr(Objective ≤ X)
[0,2450)	0
2450	0.2
2450	0.4
3180	0.6
4460	0.8
5520	1

Plan 2 Objective function outcome (X)	Pr(Objective ≤ X)
[0,2310)	0
2310	0.2
2450	0.4
3300	0.6
4520	0.8
5600	1



ii) Explain if one of the two plans (Plan1, Plan2) dominates the other deterministically or stochastically. [5]

In case of deterministic dominance, we have seen that the ranges of outcomes for alternative plans are disjoint or intersect at a single point. This is obviously not the case here as the ranges of outcomes intersect at multiple points.

For a minimization problem (students can see that this is a minimization problem because PF1 has the lowest objective value in s1 compared to any other plan), Plan A would stochastically dominate Plan B if for all possible values of x, $\Pr(Objective \leq x|Plan\ A) \geq \Pr(Objective \leq x|Plan\ B)$. This is obviously not the case here as the cumulative risk profiles cross each other.

iii) Explain which of the two plans (Plan1, Plan2) you would recommend to a planning agency that minimizes the maximum regret and comment if this criterion helps mitigate the risk of running an expensive power system. [5]

The regret formula the students were taught in class is $R(x,s) = f(x,s) - \min_x f(x,s)$. Using that formula, they can recognize that PF1 provides them with the value of

$\min_x f(x, s1)$ etc. The term $f(x, s)$ corresponds to the scenario s row of the Plan column.

Table XX. Regret in £

	Plan1	Plan 2
S1	270 (5520-5250)	350 (5600-5250)
S2	195 (4460-4265)	255 (4520-4265)
S3	30 (3180-3150)	150 (3300-3150)
S4	0 (2450-2450)	0 (2450-2450)
S5	490 (2450-1960)	350 (2310-1960)
Max regret	**490**	**350**

I would recommend plan 2 because this is the plan that minimizes the maximum regret (see table above). However, I would warn the agency than plan 2 is more expensive in the worst case scenario. Hence, it is not necessarily mitigating the risk of running an expensive power system.

b) For this part of the question assume that only scenarios S1, S3, and S5 are possible, and their total probability is 1. Using equal probabilities (1/3 for each scenario) in a stochastic program, the planning agency obtained plan SP, shown in Table 4 below. The planning agency can hire an expert, who will indicate only one scenario as plausible. The expert is correct 95% of the time.

Table 4. Objective function value in £

		Plans			
		PF1	**PF3**	**PF5**	**SP**
Scenarios	**S1**	5250	5400	6610	5520
	S3	3660	3150	3330	3180
	S5	3660	2870	1960	2450

i) Draw a decision tree that can help the planning agency calculate the expected value of the expert's imperfect information. Please explain how many nodes the tree has, their order, what each node represents and the rationale you followed to estimate the probabilities used. Please note that you do not need to enumerate decision alternatives and add numerical values for consequences. [8]

The decision tree has four chance nodes and three decision nodes:

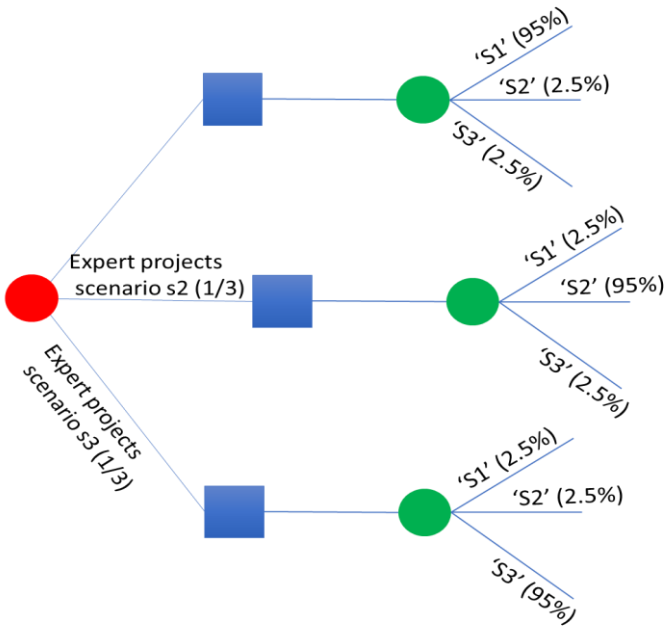
(1) The first chance node precedes the decision node(s) and reflects the uncertainty with respect to the expert's projections.

(2) Three decision nodes exist, one for each branch originating from the first chance node. This type of decision trees show that the decisions are conditional on the information shared by the expert. In other words, they recognize that the decisions happen after the expert shared their projection.

(3) One chance node following each decision node. These chance nodes reflect the uncertainty with respect to the expert's error.

Each chance node is followed by three branches, representing the three scenarios (s1, s3, s5). The three branches originating from the first chance node have equal probabilities (1/3 per scenario). However, the three branches originating from each of the second set of nodes have different probabilities: 95% probability for the scenario the expert projected (expert is

correct) and 5% probability equally split between the two scenarios that the expert did not project (expert is wrong).



ii) Using the information from Table 4, can you estimate the expected value of imperfect information, an upper bound or a lower bound of it? If yes, please provide an estimate. If no, please explain what information you are missing. [10]

To estimate the value of imperfect information, we would need to calculate the difference between the objective function we get from a stochastic program with the three equiprobable scenarios, and the sum of the probability weighted optimal objective functions at each decision node. Hence, we would need to solve three stochastic programs, one at each decision node. Given that no information has been provided to solve three stochastic programs, we are certain we cannot estimate it precisely.

However, we can estimate a bound because we already know a potentially suboptimal plan at each decision node (PF1, PF3, PF5). Hence, we can calculate a lower bound for the expected value of imperfect information.

We already have the objective function from the stochastic program
(1/3*5520+1/3*3180+1/3*2450) = 3716.67£

Expected value at top decision node if PF1 is followed: 0.95*5250+0.025*(3660+3660) = 5170.5£

Expected value at middle decision node if PF3 is followed:
0.95*3150+0.025*(5400+2870) = 3199.25£

Expected value at bottom decision node if PF5 is followed:
0.95*1960+0.025*(6610+3330) = 2110.5£

Hence the expected value at the first chance node of our decision tree will be 3493.42£

Therefore, a lower bound for the expected value of imperfect information is 3716.67-3493.42=223.25£

3. Question on modelling a new technology

According to the IPCC Climate Change 2023 report "Net zero CO2 energy systems entail in addition to a substantial reduction in overall fossil fuel use, … use of carbon capture and storage (CCS) in the remaining fossil fuel systems." Deploying CCS at a power plant or industrial facility generally entails three major steps: capture, transportation, and storage.

a) Post-combustion carbon capture is the primary method used in existing power plants and is illustrated in Figure 1 below.

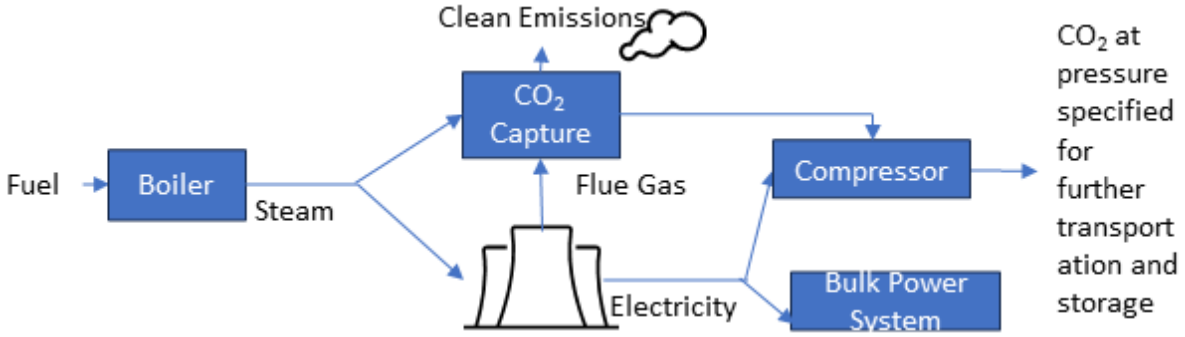


Figure 1. Simplified description of a fossil fuel power plant with carbon capture capabilities. The plant is equipped with (I) capture technology equipment that needs to use steam as input and (II) a compressor that changes the pressure of the CO_2 to delivery standards.

i) List which parameters of the basic capacity expansion model would be different for a fossil fuel power plant with Carbon Capture vs. a fossil fuel power plant without Carbon Capture. Please discuss if you expect their values to be higher, lower, or equal, and why. [8]

In the Addendum, the basic capacity expansion model has three parameters indexed by g ($AVAIL_{g,t}, FixedCost_g, VariableCost_{g,t}$). The last two parameters will be different between the two types of plants. The plant with Carbon Capture will have:

- higher fixed cost ($FixedCost_g$) because capital will need to be invested to buy the carbon capture equipment and the compressor.
- higher variable cost ($VariableCost_{g,t}$) for two reasons:
  - the efficiency will fall as some of the steam will be directed to the CO2 capture. So for the same amount of fuel burned, less electricity will be produced
  - the output to the bulk power system will reduce as some of the produced electricity will be used to operate the compressor.

ii) Please modify the basic capacity expansion model to consider a carbon tax ($PRICE^{CO2}$) and investments in natural gas power plants with Carbon Capture that can switch on/off the Carbon Capture capability (i.e., their emission rate in $\frac{tnCO_2}{MWh}$ is $\overline{em}$ when the Carbon Capture is off and $\underline{em}$ when the Carbon Capture is on). [12]

The basic capacity expansion model (provided in the Addendum) will be modified as follows:

- Sets: A new element in the set of generators G is needed for plants with Carbon Capture

- New Parameters:
  - em_g: Emission rate per plant g (units: $tnCO_2/MWh$); For the carbon capture plant, this parameter has value $\underline{em}$.
  - EM_{ccs_off}: Emission rates for plant with carbon capture (units $tnCO_2/MWh$). Its value is $\overline{em} - \underline{em}$
  - $VC_INC_{ccs_on}$: Additional variable cost when carbon capture is on (in £/MWh)
  - $PRICE^{CO2}$: Carbon tax (units £/ $tnCO_2$)
  - M: parameter with big value
- New decision variables:
  - u_t: Binary decision variable indicating if the carbon capture is on or off. 1 corresponds to on; 0 corresponds to off
  - $emCCS_t$: Nonnegative decision variable tracking emissions from the plant with carbon capture when the carbon capture is off
  - $VCCCS_t$: Nonnegative decision variable tracking the additional costs to operate the carbon capture equipment
- New constraints:
  - $-\mathrm{M} \cdot u_t \leq emCCS_t - EM_{ccs_{off}} \cdot gen_{g=NGCCS,t} \leq \mathrm{M} \cdot u_t$
  - $-\mathrm{M} \cdot (1 - u_t) \leq VCCCS_t - VC_INC_{ccs_on} \cdot gen_{g=NGCCS,t} \leq \mathrm{M} \cdot (1 - u_t)$
- New objective function term:
  - $PRICE^{CO2} \cdot \sum_g em_g \cdot gen_{g,t} + PRICE^{CO2} \cdot emCCS_t + VCCCS_t$

b) Assume that there is uncertainty around the carbon tax policy that can be described with two scenarios. In scenario 1, the carbon tax has a constant value (i.e., $PRICE^{CO2}$). In scenario 2, the carbon tax varies per hour of the year ($\sum_t PRICE_t^{CO2}$) and has the same annual average value as in scenario 1 (i.e., $PRICE^{CO2} = \frac{\sum_t PRICE_t^{CO2}}{\sum_t H_t} where\ H_t\ is\ the\ duration\ of\ its\ interval\ t$).

i) Explain if you expect CO_2 emissions in scenario 1 to be higher, lower, or equal compared to emissions in scenario 2. [6]

It depends. If the constant value is high enough for the carbon capture to be always on in scenario 1, we would expect that the emissions in scenario 1 are lower or equal to the emissions in scenario 2 because under scenario 2 there could be hours when the carbon tax is not high enough for the carbon capture to turn on.

If the constant value is so low that the carbon capture never turns on in scenario 1, we would expect the emissions to be lower in scenario 2.

ii) What is the regret of the Expected Value solution in scenario 1? [4]

The problem to yield the perfect foresight solution for Scenario 1 is identical to the problem we have to solve for the Expected Value solution because the EV problem will use for the uncertain parameter its expected value across all scenarios. .

The regret formula the students were taught in class is $R(x,s) = f(x,s) - \min_x f(x,s)$.

Given that the problems solved to yield the perfect foresight solution for scenario 1 and the EV solution are identical, the regret here is zero.

iii)Show that the equation below holds.

$$EVPI + VSS = \pi_2 \cdot Regret(EVsolution, scenario2)$$

where

- $EVPI$ is the expected value of perfect information.
- VSS is the value of the stochastic solution.
- π_2 is the probability of scenario 2.
- $Regret(EVsolution, scenario2)$ is the regret of the Expected Value solution under scenario 2. [4]

From the lecture notes, we know that $EVPI + VSS = EEV - PF$

We also know that $EEV = \pi_1 \cdot f(EVsolution, scenario1) + \pi_2 \cdot f(EVsolution, scenario2)$

$$PF = \pi_1 \cdot f(PF1, scenario1) + \pi_2 \cdot f(PF2, scenario2)$$

Having recognized in (3) that the first term in EEV and PF is the same (i.e., $f(EVsolution, scenario1) = f(PF1, scenario1)$), then we obtain

$$EEV - PF = \pi_2 \cdot f(EVsolution, scenario2) - \pi_2 \cdot f(PF2, scenario2)$$

The regret of the EV solution in scenario 2 is $f(EVsolution, scenario2) - f(PF2, scenario2)$.

Hence, we have shown that $EVPI + VSS = \pi_2 \cdot Regret(EVsolution, scenario2)$